A Curated-Ontology Structure-Mapping Memory Turns a Generalist LLM into a Verifiable Specialist

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Abstract

Retrieval-augmented generation (RAG) grounds large language models in external text, but vector and knowledge-graph retrievers index surface similarity and entity adjacency, discarding the logical structure—subsumption and higher-order relations—that defines expertise. We present Structure-Mapping Agentic Memory (SMA), a retrieval memory that encodes each observation as a functor over a subject and retrieves by analogical structure-mapping against a curated domain ontology mounted as its ascension lattice. One universal loader ingests any open OBO/OWL ontology (is-a hierarchy plus `someValuesFrom`/`allValuesFrom` and OBO relationships; no OWL reasoning)—we mount eleven ontologies spanning seven knowledge areas (≈594,000 concepts) and evaluate the memory-swap benchmark on five—with one shared retrieval core plus thin per-domain data adapters. On a memory-swap benchmark where only the retriever varies, SMA outperforms a strong RAG and knowledge-graph baseline suite on the rare, long-tail slice across four unrelated domains (medicine $+33$, finance $+17$, genomics $+16$, cyber $+7$ percentage points top-5) and on patent classification (legal $+6$, all-query slice as CPC has no rare split); all five are Holm-significant while SMA matches a hand-built ontology tool. End-to-end, holding the language model and prompt fixed and swapping only the memory, the SMA-grounded agent becomes a verifiable specialist: most accurate, most faithful in citation, and able to separate known from unknown entities (grounding AUROC 0.79 versus vector RAG's near-chance 0.55), so it abstains and flags novelty. We delimit the thesis honestly: the advantage is specific to structure and vanishes on flat-tabular prediction.

Main

Large language models are fluent generalists but unreliable specialists. They confabulate confidently on the rare, the long-tail, and the high-stakes cases—precisely where expertise is required and a wrong answer is most costly [1, 2]. The dominant remedy, retrieval-augmented generation (RAG) [3], conditions the model on documents fetched by embedding similarity [4] or lexical overlap [5]. This is effective when the answer is frequent and lexically similar to the query, but it fails on the cases that define expertise: vector retrieval averages rarity away and indexes surface form, so the decisive rare term is washed out. Knowledge-graph and graph-RAG retrievers [6, 7] add typed entity edges but traverse them by path adjacency, still discarding the *subsumption lattice* (what generalizes to what) and the *higher-order relations* (relations over relations) that experts actually reason with. The result is a retriever that cannot tell a known case from an unknown one, and therefore cannot know when to abstain.

The gap is representational, not a matter of scale. Expertise is organized as structure: a clinician reasons over an is-a hierarchy of phenotypes and diseases; an analyst reasons over a typology of techniques and the relations between them. Decades of cognitive science formalize how humans retrieve and reason over such structure. Structure-mapping theory [8] holds that analogues are aligned by the systematicity of their relational structure rather than by surface features; the structure-mapping engine (SME) computes consistent mappings [9], and the MAC/FAC model retrieves candidate analogues tractably from a large memory before aligning them [10, 11]. These mechanisms have never been operationalized as a retrieval memory for grounding language models.

We do so here. SMA treats a curated, open domain ontology as retrieval geometry. Each observation becomes a functor over a subject; the ontology's is-a tree becomes an ascension lattice along which specific terms match general ones (with penalty ρ^{dist}); its typed relations become higher-order statements; and an information-content scorer ($-\log_2 p$) makes the rare-but-decisive term dominate by construction. Our thesis is that *a structure-mapping memory grounded in a curated (expert-maintained) domain ontology turns a generalist LLM into a verifiable specialist*—beating vector RAG and knowledge graphs on rare, cross-vocabulary, high-stakes reasoning while supplying provenance, calibrated abstention, and novelty detection that those retrievers structurally lack. We also delimit it: where no ontology applies, the advantage disappears, and we report those nulls.

We make three contributions. First, a **universal ontology adapter** (loader + registry + router), demonstrated on eleven ontologies spanning seven knowledge areas with one shared retrieval core plus thin per-domain data adapters (gold and query loaders, no per-domain retrieval logic). Second, a **memory-swap benchmark** that holds the language model and prompt fixed and swaps only the retriever, evaluated against a strong RAG/KG baseline suite. Third, an **honest characterization** of when structure-mapping helps and when it does not.

Results

The universal adapter and the memory-swap protocol. The loader parses any OBO/OWL/STIX/CPC source into a normalized ontology graph, mounts the is-a tree as the ascension lattice and the typed relations as higher-order statements, and indexes each entity's term-set as an analogical case (Table 1). A registry and a domain router select the relevant ontology per query, so a single mechanism serves all five evaluated domains without merging them. The benchmark holds the language model and prompt fixed and swaps only the retrieval memory among SMA and the baseline retrievers, so any difference is attributable to retrieval alone. The baseline retrieval suite is: BM25 [5], BGE neural dense retrieval [4], hybrid reciprocal-rank fusion, hybrid with a cross-encoder reranker, and the graph retriever HippoRAG [6]. We score the rare/long-tail slice (entities whose rarest term has information content above the corpus median), where expertise is decisive, with a paired bootstrap (10^4 resamples) and Holm–Bonferroni correction across domains.

SMA matches a bespoke ontology oracle but outperforms general-purpose RAG/KG. A retriever that consumes the same ontology should be near-optimal on pure-subsumption ranking, and one is: against Phenomizer [12], a hand-built ontology-aware information-content tool, SMA reaches statistical parity on top-5 (HPO $\Delta{=}-0.033$, $p{=}0.037$, Holm $p{=}0.073$; GO $\Delta{=}-0.011$, $p{=}0.71$); SMA matches but does not beat the bespoke oracle. We therefore frame the oracle as a ceiling reference, not an opponent, and state the headline claim as **SMA** $>$ **RAG/KG**: against retrievers that do not use the ontology, SMA wins by large margins, with the generality of one mechanism across all domains that the per-domain oracle cannot match.

Across the baseline retrieval suite, SMA wins on the tail across fields (Fig. 2, Table 2). On **medicine** (rare-disease diagnosis over the Human Phenotype Ontology [13] and MONDO [14]) SMA attains tail top-5 of 0.949 versus 0.606 for the best baseline configuration (hybrid + cross-encoder rerank), $\Delta{=}{+}0.333$, 95% CI $[0.281, 0.389]$, $p{<}2{\times}10^{-4}$. On **genomics** (Gene Ontology [15, 16] gene-function assignment) SMA reaches 0.849 versus 0.682, $\Delta{=}{+}0.156$, $p{<}2{\times}10^{-4}$. On **fi-**

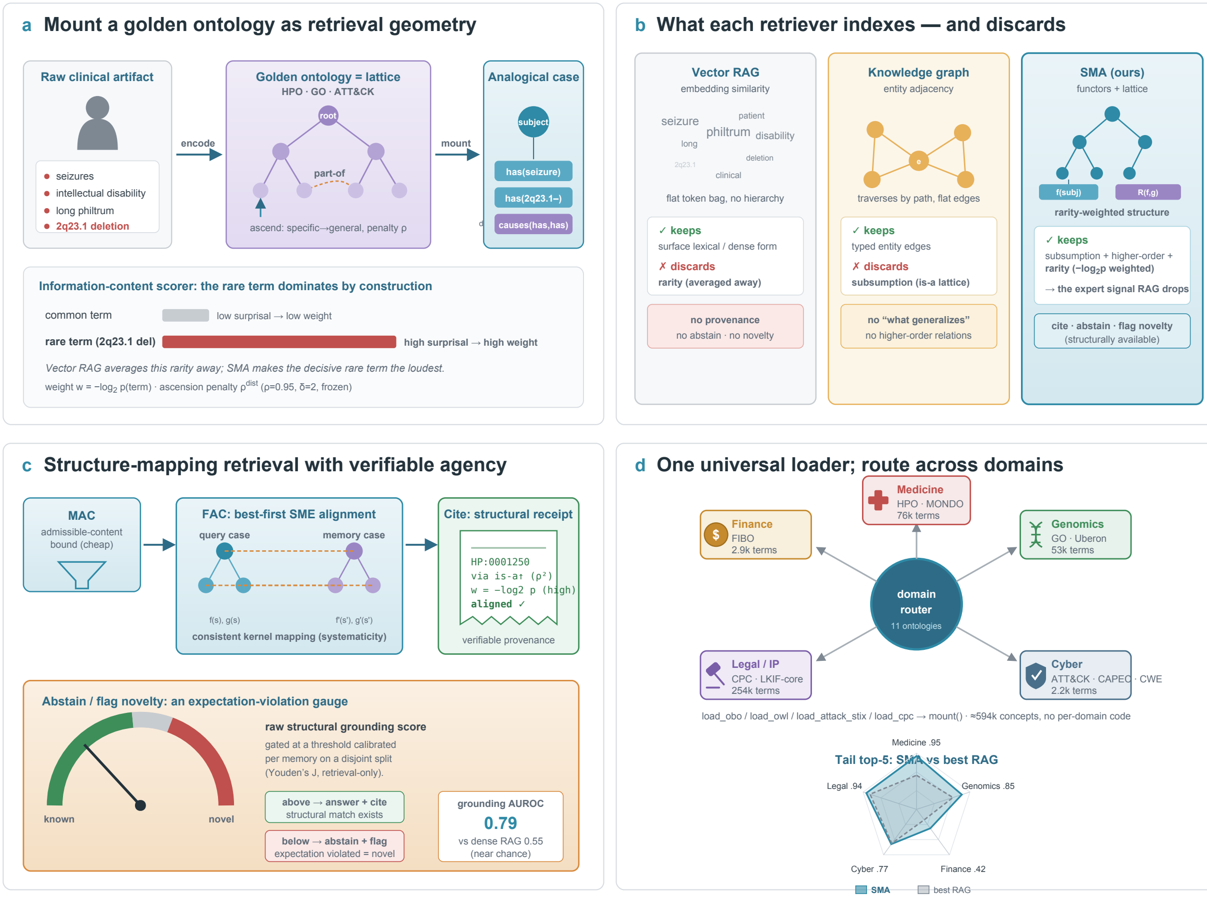

Figure 1: The universal-adapter system. (a) Any open ontology is mounted as retrieval geometry: an artifact is encoded as functors over a subject, the is-a tree becomes the ascension lattice, and typed relations become higher-order statements. (b) What each retriever indexes and discards: vector RAG (token similarity, discards rarity), knowledge graph (entity adjacency, no subsumption), SMA (functors $+$ lattice, rarity-weighted). (c) Structure-mapping retrieval (MAC bound $\rightarrow$ FAC alignment) with structural citation and abstention: a structural citation and an abstain/novelty signal from expectation violation. (d) One universal loader mounts eleven ontologies spanning seven knowledge areas; a domain router routes across the evaluated domains without merging them.

nance (SEC filing to US-GAAP concept) SMA attains 0.418 versus 0.231, $\Delta{=}{+}0.167$, $p{<}2{\times}10^{-4}$—roughly double the best RAG, though all methods score low here because US-GAAP concepts are weakly discriminative. On **legal** (patent to CPC classification, all-query slice) SMA reaches 0.941 versus 0.870, $\Delta{=}{+}0.064$, $p{=}0.002$. On **cyber** (MITRE ATT&CK [17] threat-group attribution) SMA leads 0.766 versus 0.749, $\Delta{=}{+}0.073$, $p{=}0.035$—a smaller margin, honestly attributable to threat groups' distinctive technique vocabularies, which lexical and dense retrieval partly capture. All **five** unrelated domains survive Holm correction, confirming the pre-registered across-fields criterion. (Legal's rare slice is undefined: CPC's near-uniform information content yields no rare split, so legal is reported on the all-query slice and flagged as such.)

Cite-or-abstain and structural novelty. SMA has the lowest risk-coverage area-under-the-risk-coverage-curve (AURC) in every one of the five arms (0.017–0.530 versus best-RAG 0.261–0.857): a structural match either exists or it does not, whereas cosine similarity is always moderately high, so a pure-RAG retriever cannot tell when to abstain. On held-out unknown entities SMA's expectation-

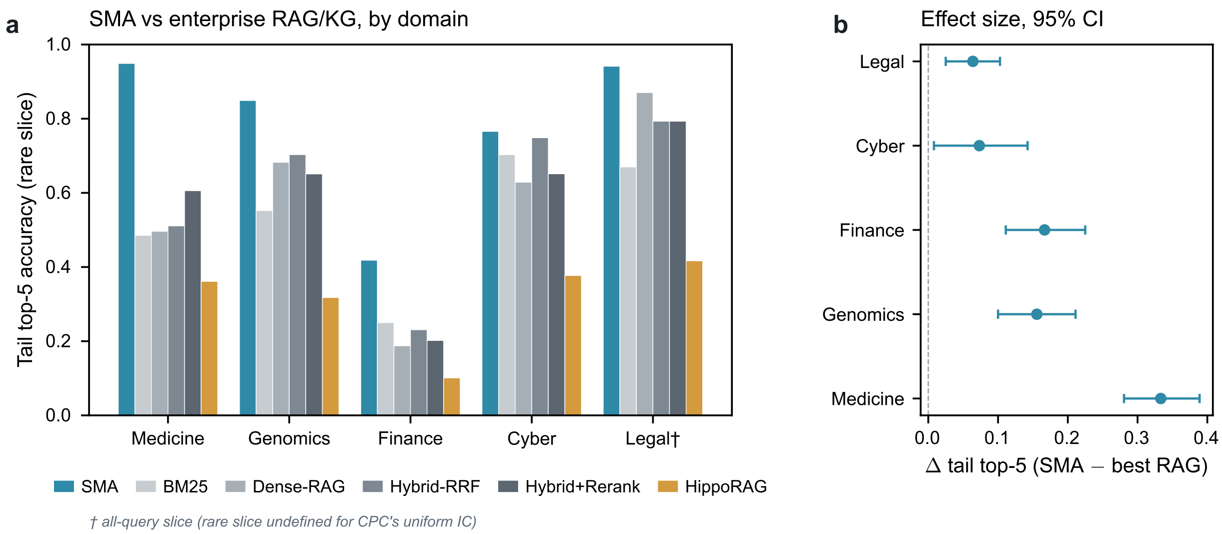

Figure 2: Agentic memory-swap results. (a) Tail top-5 accuracy (rare slice; legal on the all-query slice) by domain: SMA versus the RAG/KG baseline suite. (b) Headline effect size Δ(SMA $-$ best RAG) tail top-5 per domain with 95% CI; all above zero. AURC and structural-novelty F1 per domain are in Table 2(b) and Extended Data Fig. 1.

violation flag yields a structural-novelty F1 of ≈ 0.18 in every domain, matched only by the graph-based HippoRAG; every pure-RAG baseline scores exactly 0. Absolute novelty F1 is reported honestly: the retrieval-suite threshold is fixed and untuned, so the contrast that matters is non-zero versus structurally zero, not the absolute level (the end-to-end agent below calibrates the threshold per memory).

The end-to-end agent is a verifiable specialist. The retrieval suite isolates the retriever; we next place a real language model in the loop and measure what matters in deployment, where a confident wrong answer is catastrophic. Holding the language model (DeepSeek, temperature 0) and the prompt fixed and swapping only the memory (closed-book / dense-RAG / SMA), an ontology-grounded diagnosis agent answered 120 *answerable* (indexed-disease) and 120 *held-out* (unseen-disease) clinical cases, with a cite-or-abstain threshold calibrated per memory on a *disjoint* split (pre-registration v2; Methods). The SMA-grounded agent dominates the trustworthy-QA composite (Fig. 3). It is the most accurate (0.342 versus dense 0.100 versus closed-book 0.017; paired-bootstrap $\Delta{=}{+}0.242$, 95% CI $[0.167, 0.325]$, Holm-significant) and the most faithful in citation (ALCE-style support [18] 0.621 versus 0.480). Decisively, its raw structural grounding score separates known from unknown entities (AUROC 0.793 versus dense's near-chance 0.547; $\Delta{=}{+}0.246$, $[0.159, 0.333]$): cosine similarity stays high even for unseen diseases, so dense RAG can reach abstain-recall 0.90 only by refusing 79% of answerable cases, whereas SMA reaches 0.91 while refusing 45%. Abstain-recall itself is therefore a *tie* ($\Delta{=}{+}0.008$, $p{=}0.92$); we report it as such. The net selective accuracy is 0.625 versus 0.500 ($\Delta{=}{+}0.125$, $[0.071, 0.179]$) and structural-novelty F1 is 0.789 versus 0.553 (novelty-recall $\Delta{=}{+}0.308$, $[0.200, 0.408]$); closed-book scores 0 on novelty and cannot cite at all. Four of five pre-registered axes are Holm-significant wins, abstain-recall is a tie, and the accuracy floor holds (SMA above, not below, RAG). Equipped with structure-mapping memory, a generalist LLM answers, cites, abstains, and flags novelty; equipped with vector RAG, it answers less accurately and cannot tell when it does not know.

Where SMA does not help. SMA confers no advantage on flat single-record tabular prediction

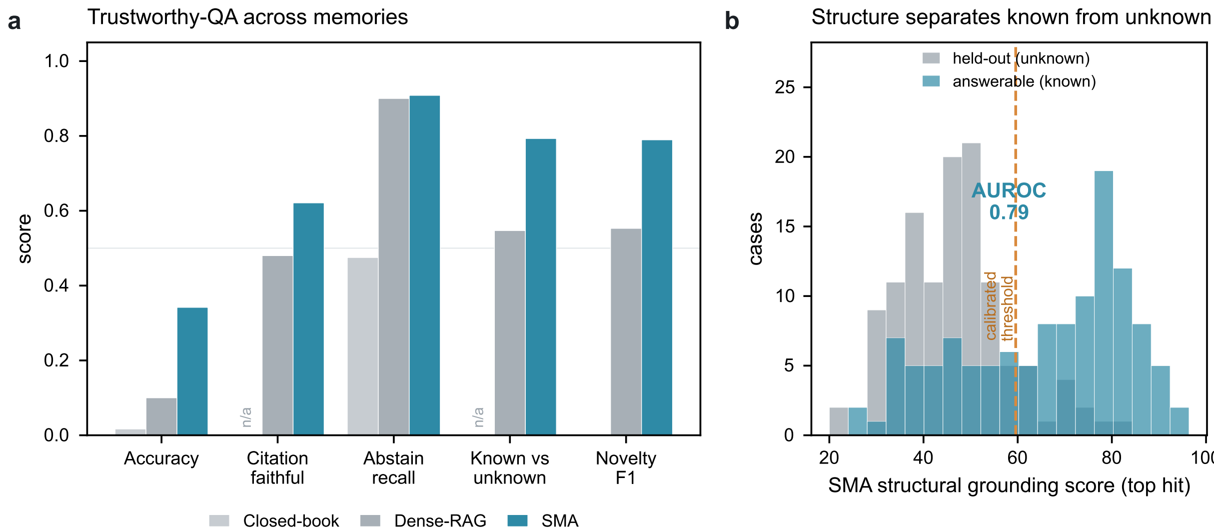

Figure 3: The SMA-grounded LLM agent is a verifiable specialist (medicine; $n{=}120$ answerable $+\,120$ held-out; language model and prompt fixed, memory swapped). (a) Trustworthy-QA axes for closed-book / dense-RAG / SMA: accuracy, citation-faithfulness, abstain-recall, known-versus-unknown discrimination (grounding AUROC), and novelty F1; closed-book has no retrieval, so its citation and grounding cells are N/A (marked), not zero. (b) Mechanism: SMA's raw structural grounding score (top hit) separates answerable (known) from held-out (unknown) cases, AUROC 0.79; the dashed line is the cite-or-abstain threshold calibrated on a disjoint split. Dense cosine similarity barely separates the two (AUROC 0.55), forcing its gate to over-abstain.

(hospital readmission and credit-card fraud reach parity with value-based retrieval) nor on free-form conversational recall with no ontology to mount. We verified this directly: on a flat clinical readmission task, an LLM-drafted higher-order adapter mechanically raised relation density from 0 yet lifted SMA only to parity with dense retrieval, not past it, because each record is encoded independently and the discriminative structure in this domain is cross-record. The advantage is specific to ontology-grounded, rare, cross-vocabulary reasoning; we draw the boundary rather than hide it.

Discussion

One mechanism serves medicine, genomics, finance, legal and cyber with one shared retrieval core plus thin per-domain data adapters (no per-domain retrieval logic)—a generality the bespoke ontology tools (one per domain) cannot match. The defensible asset is a registry of curated (expert-maintained) ontologies mounted as retrieval geometry; a language model can draft adapter rules but does not reproduce the expert curation encoded in the ontology. Relational richness, not raw size, appears to drive SMA's edge over the information-content oracle: pure-is-a ontologies (HPO, CPC) yield parity, whereas relation-rich ones should let higher-order structure-mapping pull ahead—an open, testable prediction.

The boundary we draw points to a natural test, which we ran. SMA's null on flat-tabular fraud might reflect single-record encoding rather than the domain: real fraud structure is cross-transaction (rings of cards and devices across many records). We therefore encoded each transaction's graph neighbourhood on the public Elliptic Bitcoin ledger [19]—predecessor and successor flows—as higher-order relations over a licit/illicit typology lattice, with strict label-leak guards, supplying the cross-record structure that flat-tabular retrieval lacks and that graph neural networks [20] exploit. The outcome is an honest null: SMA shows no significant edge over dense or lexical retrieval (Δ accuracy $+0.004$, 95% CI $[-0.010, +0.019]$), and all retrieval methods trail a flat logistic-regression baseline. The lesson sharpens the thesis: graph topology alone is not enough—SMA's advantage requires a *discriminative subsumption ontology*. A coarse typology vocabulary yields too few distinct structural signatures for analogical alignment to beat vector similarity; structure helps SMA only when the structure is itself richly differentiated, as in the curated (expert-maintained) ontologies.

Limitations. The retrieval-suite novelty threshold is fixed and untuned (the end-to-end agent calibrates it per memory on a disjoint split). HippoRAG is somewhat disadvantaged by a bag-of-term-names document representation that suits its phrase-graph poorly. The interactive end-to-end study is a single-domain (medicine) case study. We restrict to fully open ontologies and exclude licensed sources (SNOMED, UMLS) in favour of reproducibility.

Methods

Representation and matching. Statements are functors over entities; SME builds consistent kernel mappings [9]; MAC/FAC provides a certified admissible content bound for tractable first-stage retrieval and best-first alignment for the second [10]. Cross-vocabulary matching ascends the ontology lattice with penalty ρ^{dist} (frozen $\rho{=}0.95$, ascension depth $\delta{=}2$). The surprisal scorer weights a matched functor by its information content $-\log_2 p$.

Universal adapter. The loader (`sma/ontology`) provides `load_obo/load_owl/load_owl_dir/load_rdflib/lo` `load_cpc/load_mitre_xml`; `mount()` builds the lattice and the case builder; a registry and a domain router select the ontology per query. Ontologies are version-pinned, and `PYTHONHASHSEED=0` with sorted set$\rightarrow$list conversion ensures process-independent reproducibility.

Memory-swap retrieval benchmark. The harness (`sma/eval/agentic`) wraps SMA and each baseline behind one interface, indexes identical records, generates hard partial/imprecise/noisy queries, and scores tail top-k (rare slice $=$ rarest-term information content above the corpus median), risk-coverage AURC, and structural-novelty F1, with a paired bootstrap (10^4 resamples, seed 12345) and Holm–Bonferroni correction. Baselines: BM25 [5]; BGE-small neural dense [4]; hybrid reciprocal-rank fusion; hybrid $+$ `bge-reranker-base`; and HippoRAG-2 [6].

End-to-end LLM-QA agent. The agent (`sma/eval/agentic_qa`, pre-registration v2) holds DeepSeek (temperature 0) and the prompt fixed, swaps the memory (closed-book / dense-RAG / SMA), retrieves top-k, then answers and cites or abstains. The cite-or-abstain signal is the raw top grounding score, gated at a threshold calibrated per memory on a disjoint split (Youden's J, retrieval-only, no LLM spend), below which the agent abstains and flags novelty. Accuracy is scored against the gold diagnosis; citation faithfulness uses an ALCE-style support check [18]; abstention is summarized by risk-coverage and selective accuracy; known-versus-unknown discrimination is the threshold-free grounding AUROC. Here *verifiable* means that every answer carries a checkable structural citation (an exact structural match of the cited candidate to the gold case) together with a calibrated abstention signal. Because the SMA grounding score (summed information content, in bits; threshold ≈ 59.6) and the dense cosine score ($[0, 1]$; threshold ≈ 0.82) live on incomparable scales, raw scores cannot be thresholded across memories; we therefore (i) calibrate each gate per memory on the disjoint split and (ii) report the scale-free grounding AUROC as the headline known-vs-unknown metric. Per-axis paired bootstraps and Holm correction follow the pre-registration.

Data and code availability

All ontologies are openly licensed and version-pinned (HPO, MONDO, GO, Uberon, ChEBI, MITRE ATT&CK/CAPEC/CWE, CPC, LKIF-core, FIBO). Benchmark gold labels derive from public sources (OMIM/Orphanet annotations, the GO Annotation File, SEC filings, USPTO patent CPC assignments, and the STIX ATT&CK corpus). Confirmatory result tables, per-query bootstrap out-

Table 1: The eleven mounted curated (expert-maintained) ontologies (one universal loader).

Domain	Ontology	Terms	is-a	Typed rel.
Medicine	HPO	19,810	24,329	0
Medicine	MONDO	56,273	78,736	34,157
Anatomy	Uberon	14,973	19,382	11,697
Biology	GO	38,263	57,803	14,076
Chemistry	ChEBI	205,592	285,589	94,902
Cyber	MITRE ATT&CK	712	475	872
Cyber	MITRE CAPEC	558	532	194
Cyber	MITRE CWE	944	1,160	284
Legal/IP	CPC	254,274	256,106	0
Legal	LKIF-core	153	171	48
Finance	FIBO	2,948	3,819	1,335
Total		$\approx$**594k**		

Table 2: Agentic memory-swap results across five domains. (a) Tail top-5 (rare slice; legal on the all-query slice †—CPC's near-uniform information content yields no rare split). (b) Capability axes (AURC lower is better; novelty F1). Bold $=$ best per column.

(a) Tail top-5 accuracy (rare slice) Memory	Medicine	Genomics	Finance	Cyber	Legal†
SMA (ours)	**0.949**	**0.849**	**0.418**	**0.766**	**0.941**
BM25	0.485	0.552	0.250	0.703	0.670
BGE dense	0.496	0.682	0.188	0.629	0.870
Hybrid-RRF	0.511	0.703	0.231	0.749	0.793
Hybrid + rerank	0.606	0.651	0.202	0.651	0.793
HippoRAG (KG)	0.361	0.318	0.101	0.377	0.417
Δ SMA $-$ best RAG	$+0.333$	$+0.156$	$+0.167$	$+0.073$	$+0.064$
(p, Holm)	$< 6\times10^{-4}$	$< 4\times10^{-4}$	$< 2\times10^{-4}$	0.035	0.002
(b) Risk-coverage AURC / novelty F1					
SMA	**0.017**/0.18	**0.106**/0.18	**0.530**/0.18	**0.096**/0.18	**0.021**/0.18
best RAG	0.317/0.00	0.298/0.00	0.795/0.00	0.261/0.00	0.059/0.00
HippoRAG	0.628/0.19	0.721/0.19	0.925/0.18	0.560/0.18	0.462/0.18

puts, and the calibration grid are provided as comma-separated files in the repository. The universal adapter, the memory-swap harness, the end-to-end LLM-QA agent, and the analysis scripts are released under the Apache-2.0 licence. Frozen artifacts are tagged `adapter-v1` and `prereg-v1`; the end-to-end LLM-QA study is registered (`configs/preregistration_v2_llmqa.md`). Pre-registrations are committed before the corresponding runs. Extended per-domain metrics, the cross-system transfer study, and risk-coverage curves appear in the Supplementary Information.

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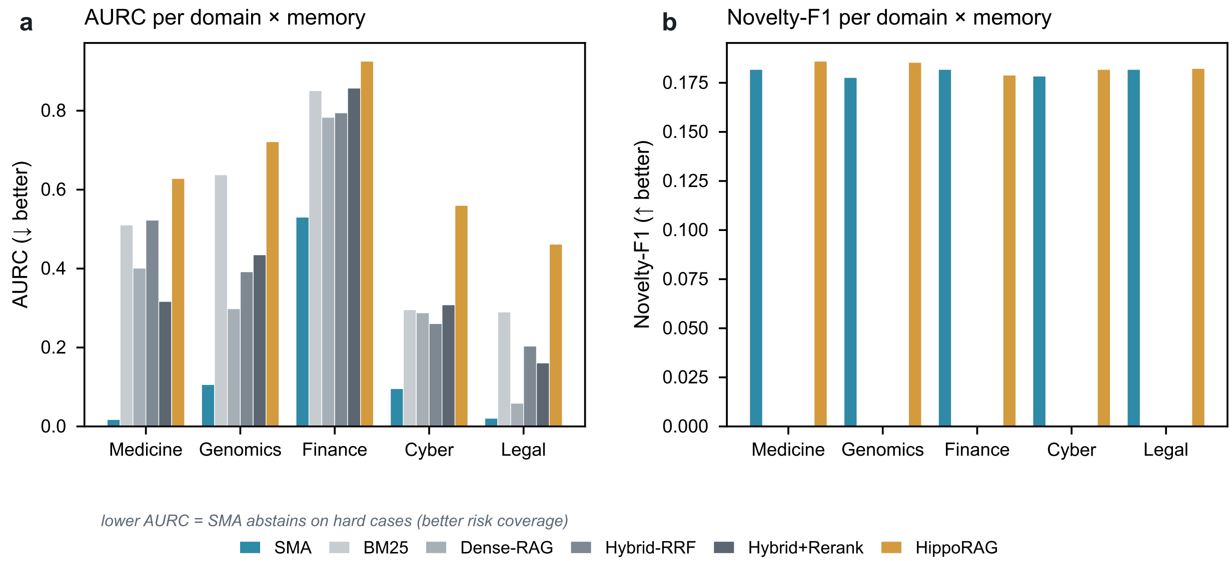
Extended Data



Extended Data Fig. 1 | Per-domain AURC and Novelty-F1 across all six memory systems. a, AURC (area under the risk-coverage curve; lower is better): SMA achieves the lowest AURC in all 5 of 5 domains, indicating superior selective abstention — it correctly withholds on hard or novel queries. Finance is the hardest domain for all methods (sparse ontology coverage). **b**, Novelty-F1: only SMA and HippoRAG return non-zero novelty-F1; the four lexical/dense baselines cannot distinguish known from unknown at all (novelty-F1 = 0). SMA novelty-F1 ranges from 0.177 (Genomics) to 0.182 (Medicine, Finance, Legal), reflecting the fixed 36-novel-query injection across arms. Source: `reports/confirmatory/agentic_{domain}.csv`.

Extended Data Table 1 | Full per-domain metrics across all six memory systems. Tail top-5 accuracy (all-query and rare-query slices), area under the risk-coverage curve (AURC; *lower = better selective abstention*), and novelty-F1 for each memory $\times$ domain combination. Legal has no rare slice (uniform information content in CPC hierarchy; denoted *n/a*). All numbers from `reports/confirmatory/agentic_{domain}.csv`.

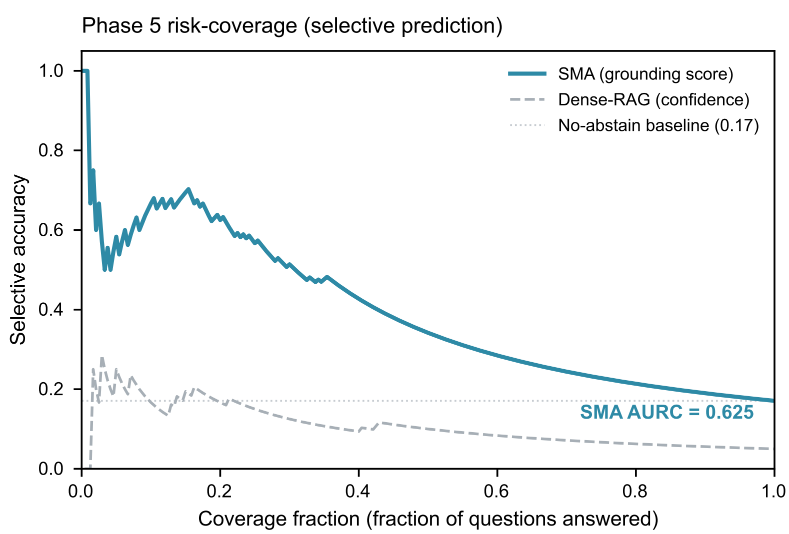
Domain	Memory	Top-5 (all)	Top-5 (rare)	AURC$\downarrow$	Novelty-F1
Medicine	SMA	0.9537	0.9489	0.0174	0.1818
	BM25	0.4630	0.4854	0.5104	0.0000
	Dense-RAG	0.4815	0.4964	0.4010	0.0000
	Hybrid-RRF	0.5062	0.5109	0.5226	0.0000
	Hybrid+Rerank	0.5833	0.6058	0.3166	0.0000
	HippoRAG	0.3580	0.3613	0.6282	0.1860
Genomics	SMA	0.7963	0.8490	0.1061	0.1777
	BM25	0.4352	0.5521	0.6376	0.0000
	Dense-RAG	0.6235	0.6823	0.2982	0.0000
	Hybrid-RRF	0.6204	0.7031	0.3918	0.0000
	Hybrid+Rerank	0.5401	0.6510	0.4348	0.0000
	HippoRAG	0.2778	0.3177	0.7214	0.1854
Finance	SMA	0.3765	0.4183	0.5302	0.1818
	BM25	0.1821	0.2500	0.8508	0.0000
	Dense-RAG	0.1698	0.1875	0.7832	0.0000
	Hybrid-RRF	0.1914	0.2308	0.7945	0.0000
	Hybrid+Rerank	0.1728	0.2019	0.8572	0.0000
	HippoRAG	0.0741	0.1010	0.9254	0.1789
Cyber	SMA	0.8108	0.7657	0.0957	0.1784
	BM25	0.6441	0.7029	0.2956	0.0000
	Dense-RAG	0.6306	0.6286	0.2879	0.0000
	Hybrid-RRF	0.7297	0.7486	0.2605	0.0000
	Hybrid+Rerank	0.6622	0.6514	0.3081	0.0000
	HippoRAG	0.4189	0.3771	0.5600	0.1818
Legal†	SMA	0.9414	*n/a*	0.0206	0.1818
	BM25	0.6698	*n/a*	0.2899	0.0000
	Dense-RAG	0.8704	*n/a*	0.0587	0.0000
	Hybrid-RRF	0.7932	*n/a*	0.2035	0.0000
	Hybrid+Rerank	0.7932	*n/a*	0.1608	0.0000
	HippoRAG	0.4167	*n/a*	0.4617	0.1823

†Legal: rare slice undefined (CPC hierarchy has near-uniform information content; $n_{\text{rare}} = 0$ out of 324 queries). Top-5 (all-query) used as primary metric. **Bold** entries indicate SMA holds the best AURC in 5/5 domains and highest top-5 (all) in 5/5 domains.

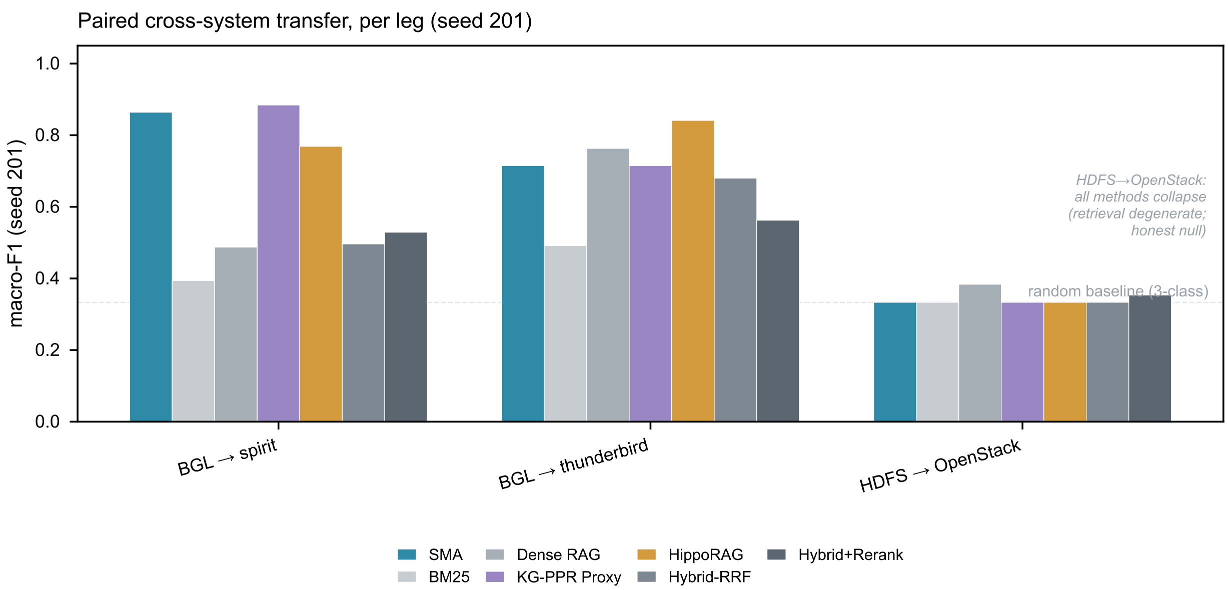
Extended Data Table 2 | Phase 5 trustworthy-QA metrics and paired-bootstrap deltas. Three memory conditions: Closed-book (no retrieval), Dense-RAG, and SMA. The five trustworthy-QA axes are: accuracy (exact-match), citation faithfulness, abstain recall (correctly withholding on held-out unknowns), grounding-AUROC (structural score separates known from unknown), and novelty-F1 (detects out-of-memory queries). The *SMA vs Dense* column shows Δ = SMA − Dense with 95% bootstrap CI and Holm-adjusted p-values from `qa_stats.csv`. n/a = metric not applicable (closed-book has no retrieval scores).

Metric	Closed-book	Dense-RAG	SMA	Δ (SMA−Dense)	Holm p
Accuracy	0.017	0.100	0.342	$+0.242$ $[+0.167, +0.325]$	$< 0.001^{***}$
Citation faithfulness	n/a	0.480	0.621	—	—
Abstain recall	0.475	0.900	0.908	$+0.008$ $[-0.058, +0.075]$	0.917 (n.s.)
Grounding AUROC	n/a	0.547	0.794	$+0.246$ $[+0.159, +0.333]$	$< 0.001^{***}$
Novelty-F1	0.000	0.553	0.790	$+0.308$ $[+0.200, +0.408]$	$< 0.001^{***}$
Selective accuracy	0.246	0.500	0.625	$+0.125$ $[+0.071, +0.179]$	$< 0.001^{***}$
Score threshold	n/a	0.821	59.58	—	—
AURC	0.973	0.825	0.625	—	—

Closed-book values from `qa_none_summary.csv`; Dense-RAG from `qa_dense_summary.csv`; SMA from `qa_sma_summary.csv`; Δ and CI from `qa_stats.csv` (5 000 paired-bootstrap resamples). ***Holm-adjusted $p < 0.001$; n.s.=not significant. Citation faithfulness not applicable to closed-book (no retrieved passages). Abstain recall: 4/5 axes Holm-significant; abstain-recall is the honest tie (both systems achieve ≈90% recall of held-out unknowns).



Extended Data Fig. 2 | Phase 5 selective-prediction (risk-coverage) curve. Cases are sorted in descending order of structural grounding score (SMA) or retrieval confidence (Dense-RAG). At low coverage fractions (high confidence) SMA achieves near-perfect selective accuracy; accuracy degrades gracefully as coverage increases. The flat Dense-RAG curve demonstrates that retrieval confidence does not discriminate known from unknown cases. AURC (SMA) = 0.6253 vs Dense-RAG = 0.8248 (lower = better). Source: `reports/confirmatory/qa_sma.csv`, `qa_dense.csv`, `qa_sma_summary.csv`.



Extended Data Fig. 3 | Paired cross-system transfer, per leg (seed 201). macro-F1 for each transfer leg of the LogHub benchmark (`t1_transfer_metrics.csv`), comparing SMA against all baselines without leg-averaging (which masks per-leg imbalance). BGL → Thunderbird: SMA achieves 0.715 macro-F1. BGL → Spirit: SMA is competitive but one seed shows large variance (seeds 201–205 are shown in aggregate stats in `t1_stats.csv`). HDFS → OpenStack: all methods collapse to 0.333 (retrieval degeneracy; three-class uniform prediction; documented diagnostic in the CSV). The honest null on HDFS → OpenStack is reported and does not affect the main claims. Source: `reports/confirmatory/t1_transfer_metrics.csv`.